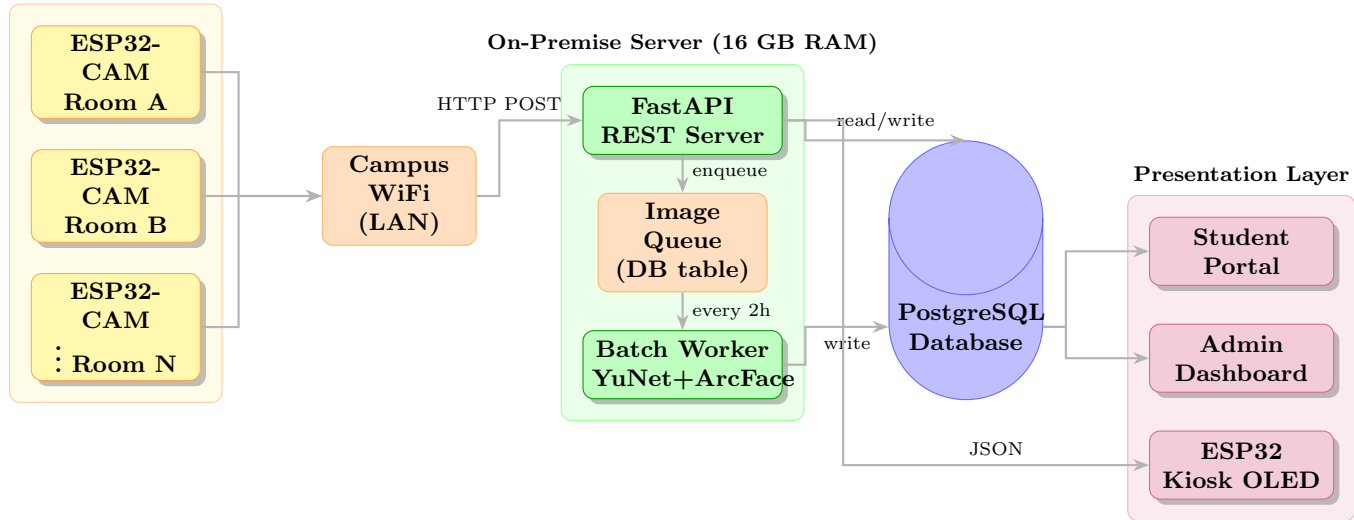
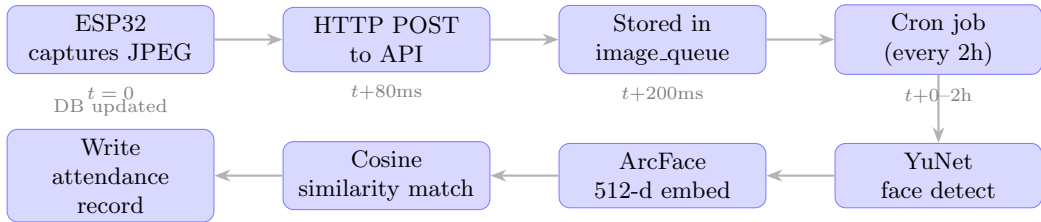


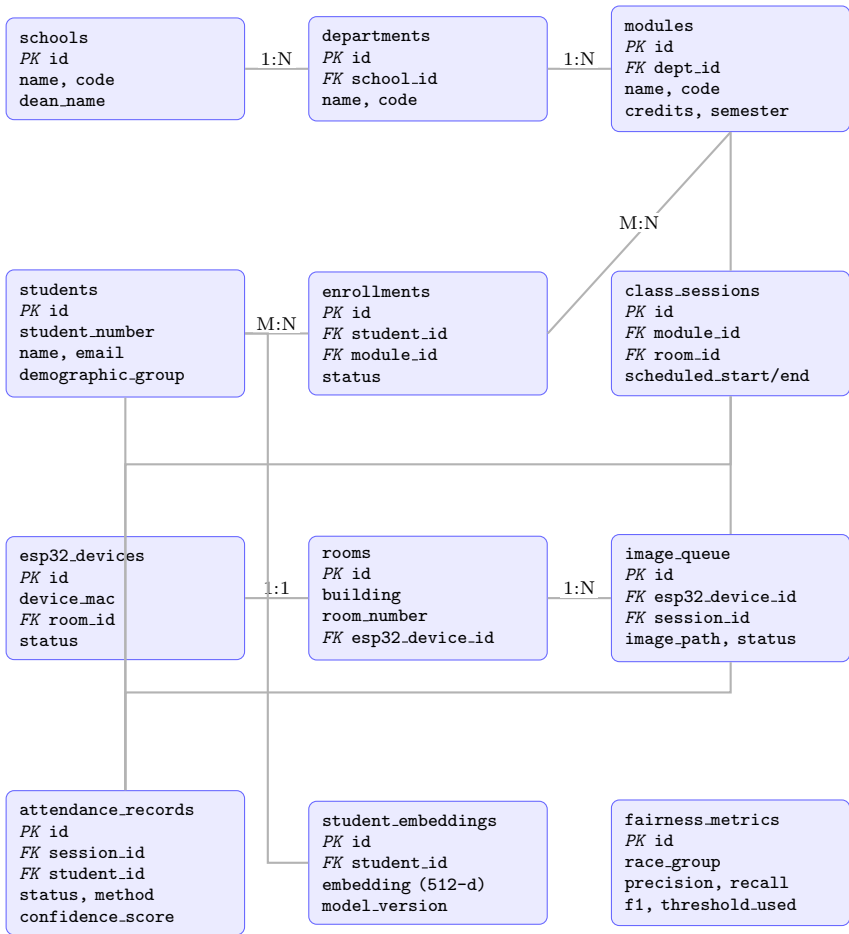
IoT Capture Layer



Batch Processing Pipeline



Entity-Relationship Diagram



Classroom Kiosk Unit

Smart Attendance

Please look at camera

Room: LT-204 | 09:03 AM

Module: CS301 | Machine Learning



OV5640 5MP Autofocus



Present



Unknown



Processing

ESP32-CAM — WiFi: campus-net

Firmware v1.0.3 — IP: 192.168.1.45

AI Attendance Node

Overview

Attendance Log

Manual Override

Devices

Fairness KPIs

94.2%

Recognition Accuracy

12847

Attendance Events

23

Manual Overrides

847

Queue Pending

198

Active Devices

Live Attendance Log

Student	Module	Time	Room	Method	Conf	Status
Alice Johnson	CS301	09:00	LT-204	Face Recog	0.87	Present
Bob Patel	CS301	09:00	LT-204	Face Recog	0.82	Present
Clara Wu	CS301	09:00	LT-204	—	—	Absent
David Osei	MA201	10:00	RM-101	Face Recog	0.91	Present
Manual Override	CS302	11:00	Fairness F1 Scores (FairFace)			— Present

Student Number

Module

Session Date

Status

Save Override

White

0.96

East Asian

0.94

South Asian

0.91

Black

0.88

Latino

0.87

SE Asian

0.85

Middle Eastern

0.84

Server Health

CPU

34%

RAM

6.2 GB

Queue Depth

847

Throughput

1240 img/hr

Uptime

14d 6h

Smart Attendance — Student Portal

Welcome, Alice Johnson

- Dashboard
- My Attendance
- Timetable
- My Modules
- Profile

87%

Overall Attendance

6

Modules Enrolled

1

At Risk

23

Sessions This Sem

Attendance by Module					
Code	Module Name	Sessions	Attended	%	Status
CS301	Machine Learning	12	11	92%	OK
CS302	Data Structures	10	9	90%	OK
CS303	Networks	8	5	63%	AT RISK
CS304	Databases	11	10	91%	OK
MA201	Statistics	9	8	89%	OK
Recent Sessions					
Date	Time	Room	Status	Method	
2026-06-20	09:00	LT-204	Present	Face Recognition	
2026-06-17	09:00	LT-204	Absent	—	
2026-06-13	09:00	LT-204	Absent	—	
2026-06-10	09:00	LT-204	Present	Face Recognition	

Attendance updated every 12 hours. Last update: Today 06:00. Contact lecturer to dispute.